Study Documents

Petition or Protest? Activism and Prosocial Behaviour

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Part I

Ethics Reflection

Study's contributions to existing research and the body of knowledge Activism can be effective for shifting societal norms (e.g., MeToo, Black Lives Matter, Fridays for Future). Experimental research has explored how effective normative and non-normative activism is at shifting support for and contribution to causes. There is some evidence that more extreme activism is typically less effective. However, this research is largely vignette-based (Feinberg et al., 2020; Thomas and Louis, 2014), and none being done within the context of a public goods game. So, the evidence is mixed. Extreme action not effective, but impacted by how violent, disruptive, or how justified the cause is.

How will participants be recruited for this study? A total of 60 participants will be recruited via WU Labs through ORSEE. Greiner (2015)

Please describe the basic research design and procedures of your study. We conduct a repeated public goods game. The baseline part of the experiment (Part 1) sets the baseline level of contribution. In a first stage, players perform a real effort task (counting 1s in a grid), earning a “wage" corresponding to their output. They then decide how much of their wage to contribute to the public good across seven repetitions.

In the treatment part of the experiment (Part 2), we use a between-player design. The participant group is randomly assigned to one of three kinds of “activism". Before the effort task, each participant is invited (simultaneously) to participate in activism. As in the baseline, in the treatment stage, players choose how much they want to contribute to the public good. Part 2 is repeated seven times. We define activism as a public, sometimes costly, voluntary signal of support for a cause, which in our game is supporting cooperation on the public good. The main distinctions between the three types are the personal costs involved and the costs imposed on others:

1. Petition: signal
2. Demonstration: signal + personal costs
3. Road blocking: signal + personal costs + cost on others

Each player goes through the control rounds (Part 1), followed by seven rounds of one of the three activism types (Part 2). The public good groups are fixed across all fourteen rounds.

Are participants compensated for participating in this study, and if so, how? WU lab participants will receive standard WU Behavioral Lab compensation (approximately €10/hour). There is a show up fee of €5, and for our experiment, the remaining compensation will be based on performance and randomly selected from one round of the experiment. Our experiment will take place in a session with multiple experiments of which one is chosen randomly for payment.

What are potential risks and costs of your experiment to participants and others, and how do you mitigate them? Risks are minimal and comparable to everyday exposure to that typically encountered in day-to-day experiences. Engaging in activism may have negative psychological effects, which would likely be equivalent to what would happen with punishment in a public goods game. To mitigate potential risks and costs, all forms of activism will invoke minimal costs to self and others, relative to overall earnings.

Is participation informed and voluntary? Yes. Participants who participate in the study will be recruited through the WU Lab study pool. All participants within this pool provide consent when they sign up to participate. Participation is voluntary, and participants may withdraw at any time throughout the study, and a guaranteed a show up fee of 5 euros for attending the session.

Are you using any deception? No, no deception will be used within the study.

Part II

Pre-Analysis Plan

1) Has any data been collected for this study already?

No, data has not been collected for this study yet.

2) Describe the key dependent variables specifying how they will be measured.

We pre-register three primary outcome variables at the individual level:

1. **Relative contribution:** the share of a participant's earnings contributed to the public good in each period:
$$q_{it} = \frac{g_{it}}{y_{it}}$$
where g_{it} denotes participant i's contribution in period t and y_{it} denotes earnings.
2. **Activism turnout:** $a_{it} \in \{0, 1\}$, equal to one if participant i engages in activism in period t, and zero otherwise.
3. **Payoff:** individual earnings, Π_{it}, in a single round t, after contribution and activism decisions.

Note: At period t, q_{it} measures willingness to contribute independently of earnings, while Π_{it} captures the material consequence of all decisions combined. For relative contribution and turnout, we calculate:

- the mean across all periods,
- the mean in the final period, and
- changes over time.

Final payoffs are determined by random selection of one of the 14 rounds. Under this payment protocol, the expected payoff equals the average earnings across rounds, giving participants an incentive to treat each round as if it were payoff-relevant. We calculate average welfare at the group level and compare welfare across treatments and baseline rounds.

3) What is the main question being asked or hypothesis being tested in this study?

This study investigates how different forms of activism affect prosocial behaviour in a repeated public goods game. In particular, we test whether activism that imposes only private costs versus activism that additionally imposes external costs on others changes:

1. contributions to the public good,

2. participation in activism, and
3. group welfare.

Our main hypotheses are:

- Relative contributions will be lowest in the roadblocking treatment and highest in the demonstration treatment:
$$\bar{q}_{RB} < \bar{q}_{BASE} \leq \bar{q}_{PET} < \bar{q}_{DEMO}$$
- Participation in activism will decrease as the costs of activism increase, at both the individual and group level:
$$\bar{a}_{RB} < \bar{a}_{DEMO} < \bar{a}_{PET} \qquad \text{(H2a: individual)}$$
$$\bar{A}_{RB} < \bar{A}_{DEMO} < \bar{A}_{PET} \qquad \text{(H2b: group)}$$
where $\bar{a}_k$ denotes the average individual probability of engaging in activism in treatment k, and $\bar{A}_k$ denotes the average number of activists per group in treatment k (range 0–4).
- Welfare will be lowest in the roadblocking treatment due to the external costs imposed on other participants:
$$\bar{W}_{RB} < \bar{W}_{DEMO} < \bar{W}_{BASE} \leq \bar{W}_{PET}$$

4) How many and which conditions will participants be assigned to?

Participants are assigned to fixed groups of four participants and play a finitely repeated public goods game. The experiment consists of a baseline phase followed by one of three treatment conditions. The conditions are:

1. **Baseline (control period):** standard repeated public goods game without activism opportunities.
2. **Petition treatment:** participants can engage in symbolic activism with no costs.
3. **Demonstration treatment:** participants can engage in activism that imposes private costs through lost earning time.
4. **Roadblocking treatment:** participants can engage in activism that imposes both private costs and external costs on other group members by reducing others' earning time.

Groups are randomly assigned to one of the three treatment conditions after the baseline rounds. Due to time constraints, the baseline will be repeated 7 times, and the treatment rounds 7 times.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

1) We first compare treatment means using ANOVAs and estimated marginal means to compare group level differences between treatments. For the main analyses, we estimate OLS regressions

with standard errors clustered at the group level. As a robustness check, we estimate generalized linear regression models to account for the bounded nature of the outcome. The main specification examines relative contributions as a function of treatment indicators and period fixed effects:

$$q_{it} = \beta_0 + \beta_1 PET_i + \beta_2 DEMO_i + \beta_3 RB_i + \gamma_t + \varepsilon_{it}$$

where γ_t denotes period fixed effects. For robustness checks we include controls such as age, gender, education.

2) We additionally estimate linear probability models examining activism participation:

$$Activist_{it} = \alpha + \beta_1 \overline{q}_{-i,t-1} + \beta_2 NegDev_{it-1} + \beta_3 PosDev_{it-1} + \beta_4 Treatment_i + \varepsilon_{it}$$

where $Activist_{it}$ is a binary indicator equal to one if participant i engages in activism in period t, $\overline{q}_{-i,t}$ denotes the average contribution of the other group members at period $t-1$, Following Fehr and Gächter (2000) we expect different activism behaviour for positive and negative deviations from the mean contribution. Thus, $NegDev_{it}$ denotes the absolute negative deviation from the group average contribution at period $t-1$, and $PosDev_{it}$ denotes the absolute positive deviation from the group average contribution at period $t-1$. Note, this regression can only be estimated from period $t = 2$ onwards, for the activism rounds. For activism participation, we estimate a linear probability model using OLS and Probit as robustness, again with errors clustered at the group level.

3) We also compare:

- contribution of activists and non-activists, and
- treatment differences in welfare and free-riding.

Finally, we analyze how contributions and turnout evolve over time.

6) Describe exactly how outliers will be defined and handled, and your precise rules for excluding observations.

We will not systematically exclude observations, however, we will conduct our robustness checks excluding behavioural outliers to test for robustness (e.g., people who don't participate in the RET across all rounds).

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

The experiment consists of two laboratory sessions with 28 participants each, resulting in a expected total sample of 56 participants. Participants are assigned to fixed groups of four participants, yielding 14 groups in total. Groups are randomly assigned by cycled allocation to one of the three treatment conditions, resulting in five groups for two treatment conditions and four groups for one treatment.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

This pre-registration is a pilot study, as a part of the Experimental Methods in Business and Economics I+II at WU Vienna. We will use this study to inform power-analyses and the design of a full-scale study which will take place in the second half of 2026, subject to funding. This pilot will also indicate if it is necessary to extend the rounds beyond seven, or if behaviour stabilises by round 5 in the treatment, and fewer repetitions can be used.

We will perform robustness checks accounting for:

- any groups who do not finish the experiment for any reason
- the treatment rounds only, for comparability with the baseline
- age, gender, education (Questionnaire)
- manipulation questions (Questionnaire)
- trust, social identification, emotional connection (Questionnaire)

Part III

Power Analysis & Budget

1 Power analysis

Table 1: Required sample size per group at $\alpha = 0.05$, power = 0.80, one-sided test.

Hypothesis	Contrast	p_1	p_2	n per group
H1: Relative contributions				
	BASE vs DEMO	0.250	0.355	236
	BASE vs RB ($\bar{q}_{RB} = 0.20$)	0.250	0.200	862
	BASE vs RB ($\bar{q}_{RB} = 0.24$)	0.250	0.240	22,872
H2a: Individual activism turnout				
	PET vs RB	0.500	0.053	12
	PET vs DEMO ($\bar{a}_{DEMO} = 0.40$)	0.500	0.400	305
	PET vs DEMO ($\bar{a}_{DEMO} = 0.10$)	0.500	0.100	16

Notes: p_1 denotes the higher proportion. Sample sizes are the minimum number of observations per group required to detect the difference between p_1 and p_2 with 80% power at a 5% significance level. BASE vs PET is omitted as both arms share the same assumed mean ($\bar{q} = 0.25$).

2 Budget estimation

We have a total of 30 euros per participant, including a 5 euro show-up fee. Thus, 25 euros can be allocated on average per participant based on the public goods game. We use the payoff scheme from Fehr and Gächter (2000):

$$\pi_i = y - q_i + \alpha \sum_{j=1}^{n} q_j, \quad \text{where} \quad 0 < \alpha < 1 < n\alpha. \tag{1}$$

Individuals can earn money based on their performance in a task and then contribute to a public good. Individual payoffs are determined by individual earnings and group contributions to the public good. We randomly draw one round from the 14 rounds each individual plays. Thus, the probability that the baseline round is selected is 0.50. Since we have three treatments, the probability that one treatment is chosen is 0.17 in total.

For the budget estimation, we make the following assumptions:

- Individuals solve no more than 12 tasks within 45 seconds. This corresponds to 3.75 seconds per task. We assume that individuals always try to maximise income, although we acknowledge that performance likely declines over time due to cognitive constraints. We can accordingly calculate the maximum income for 35 and 40 seconds in the activism treatments.
- Following Nikiforakis (2008), we assume that in the baseline period, individuals contribute on average 25% of their earnings to the public good.

- We assume that the petition treatment is similar to a cheap-talk setting. Based on the literature, we do not expect differences in relative contributions in the petition treatment (Bochet et al., 2006).
- For the demonstration treatment, we assume it is closest to the chatroom treatment in Bochet et al. (2006). Demonstration represents a form of computer-mediated communication. The chatroom treatment increased contributions by 42%. Thus, we assume a contribution rate of 35.5% in our demonstration treatment. Unfortunately, there is no data on the frequency of chatroom usage. Therefore, we make a best guess and assume that the probability of demonstration lies between 10% and 40% to remain consistent with our hypothesis. For the budget estimation, we use the midpoint of 25%.
- We assume that the roadblocking treatment is similar to a punishment–counter-punishment (PCP) setting as in Nikiforakis (2008). Thus, we use the frequency of PCP from Nikiforakis (2008) as the expected probability for roadblocking (5.3%). We assume an average relative contribution of 20% in line with our hypothesis to be lower than in baseline.
- We assume that the contribution rates stated for the treatments are average group-level rates, accounting for periods without activism.
- We assume $\alpha = 0.4$

In the following, we calculate the average payoff per treatment before deriving the total average payoff per participant assuming an earning of 10 points per correctly solved task, where 1 point = 0.15 euro.

Wages

Each player earns a wage equal to the maximum number of tasks multiplied by the points per task (10 points). Where no treatment-specific wage is assigned, the baseline wage applies:

$$w^{\mathrm{BASE}} = w^{\mathrm{PET}} = w_{na}^{\mathrm{DEMO}} = 120, \qquad w_a^{\mathrm{DEMO}} = w_a^{\mathrm{RB}} = 93.\overline{3}, \qquad w_{na}^{\mathrm{RB}} = 106.\overline{6}.$$

Group Composition

We assume activism is assigned independently with probability a, the number of activists k in a group of 4 follows a Binomial(4, a) distribution:

$$P(k;\, a) = \binom{4}{k} a^k (1-a)^{4-k}.$$

TLDR; in groups of four there can be many combinations of players which, for example, there are two activists in the group. We need to account for these many combinations, as well as removing double counting (A & B being activists in the same as B& A being activists).

For a group with k activists, the total group endowment is

$$W(k) = k\, w_a + (4-k)\, w_{na}.$$

Earnings Formula

A player retains $(1-q)$ of their own wage and receives the group project bonus $\text{MPCR} \times q \times W(k)$:

$$\pi_r(k) = (1-q)\, w_r + \text{MPCR} \times q \times W(k), \qquad r \in \{a,\, na\}.$$

The expected earnings per player, averaging over group composition, are:

$$\mathbb{E}[\pi] = \sum_{k=0}^{4} P(k;\, a) \left[\frac{k}{4}\, \pi_a(k) + \frac{4-k}{4}\, \pi_{na}(k) \right].$$

2.1 Expected Payoffs by Treatment (see excel)

Applying the earnings formula and, for DEMO and RB, weighting over group composition probabilities yields the expected payoffs reported in Table 2.

Treatment	$\bar{q}$	$\bar{a}$	$\mathbb{E}[\pi]$
BASE	0.25	—	138.00
PET	0.25	0.50	138.00
DEMO	0.355	0.25	137.47
RB	0.20	0.05	118.72

Table 2: Expected earnings per player by treatment.

2.2 Total Expected Payoff per Participant

The overall expected payoff is

$$[E(Y_i) = 0.50\, \pi_{\text{BASE}} + 0.167\, (\pi_{\text{PET}} + \mathbb{E}[\pi_{\text{DEMO}}] + \mathbb{E}[\pi_{\text{RB}}])] \times 0.15 \text{ euro per point} = 20.24 \text{ euro}$$

2.3 Maximum Payoff per Participant

If all group members contribute the (max) wage:

$$120 - 120 + 0.4 \times 4 \times 120 = 192 \implies 192 \times 0.15 \text{ euro} = 28.8 \text{ euro}$$

Part IV

Experimenter Script

Experimental Protocol

Setup

1. (30 mins before exp.) Open oTreehub/sites. Open experiment-a. Check Server settings are at STUDY, hides debug, and the admin password is set. Launch activism-v3.
2. Go to https://experiment-a.on.otree.org/demo. Enter admin password. Go to Rooms.
3. (Test day) Create in the "test" room a session with 16 participants (one of each of the three treatments will be active.) (Pilot day: Session 1: Create in s1 a session with 28 participants. Session 2: Create in s2 a session with 28 participants)
4. Copy links to computers. ***Copy site link to students not participating in the exp*** Ensure all 28 computer stations are loaded to the initial oTree session URL and remaining have access to the internet.

During

5. Welcome the participants and announce: *"Welcome, and thank you for participating in today's experiment. From this point forward, please (ensure) your mobile phones remain switched off and do not communicate with any other participants"*
6. Hand out the printed instructions to the participants: *"Please read the instructions carefully. If there are any questions at this point, please ask."*
7. Once all questions are answered, announce the beginning of the experiment: *"The experiment is now beginning. Please follow the screen. "*
8. Monitor the oTree dashboard until all 28 participants have finished the experiment. Ensuring no one is delayed on a wait page due to a technical error.

Ending

9. Open the payment page of the session. Prepare money box and participant list to sign.
10. Wait for all participants to complete the survey.
11. Announce the conclusion: *"As mentioned in the instructions, you have participated in several experiments today. The relevant experiment was "". Please remain seated quietly. I will call you up one by one by your seat number to privately provide your payment and receipt."*
12. Use the oTree admin dashboard to verify the final payouts, privately pay participants, and dismiss them.

References

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